

17	A	Using $d \sin \theta = n\lambda$, $d = \frac{n\lambda}{\sin \theta}$ $= \frac{2(500 \times 10^{-9})}{\sin 30^\circ}$ $= 2.0 \times 10^{-6} \text{ m}$ Number of diffraction lines per metre of the grating $= \frac{1}{2.0 \times 10^{-6}}$ $= 500000$ Number of diffraction lines per millimetre of the grating $= 500$
18	A	